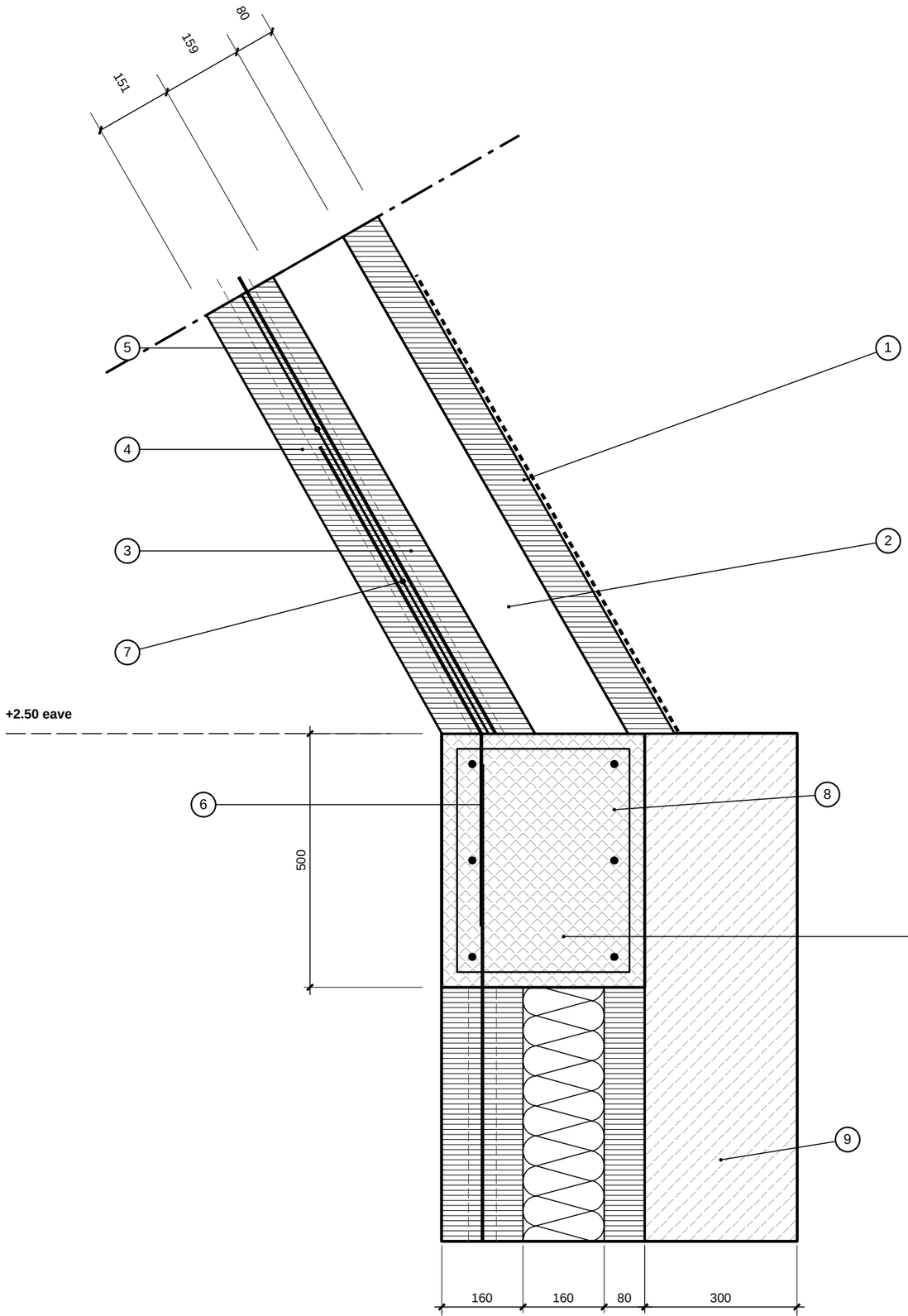
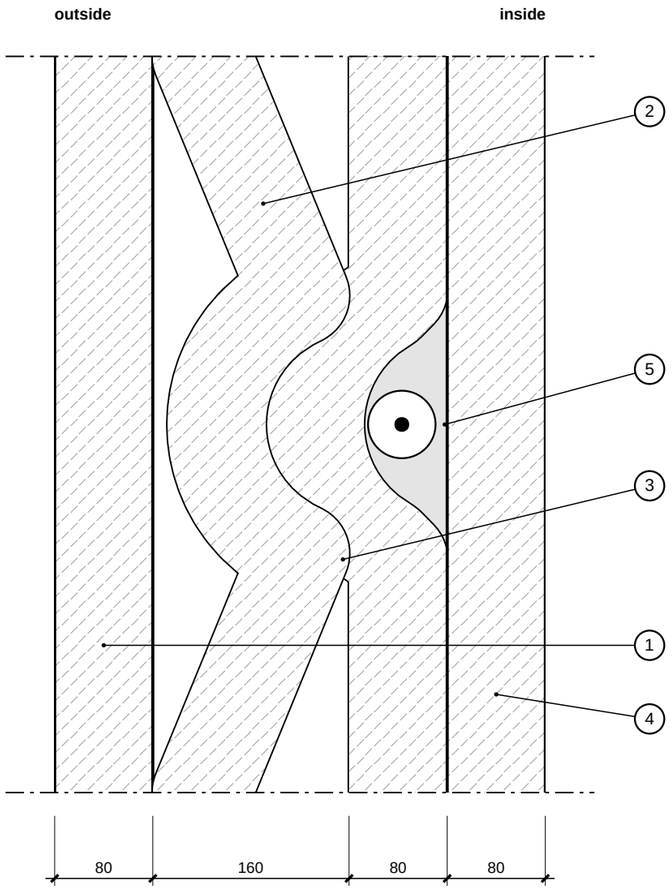


Plan of one roof piece, 1:5

layer pattern, bars every ≈ 600 along the path



Detail D1: eave, roof to wall

at the mid-span buttress, section A-A, 1:10 · piece plan 1:5

1. Roof outer layer 80, printed with the rest of the piece; cladding on the outside face (dashed line).
2. Zig-zag bead 80 in a 160 zone; it touches the outer layer and layer 2 every ≈ 600; the cells between are open air pockets for ventilation.
3. Layer 2, 80: printed bead that curves to leave a pocket for the bar.
4. Inner layer 1, 80, room side. Layers 1 and 2 together are the structural leaf: 160 at the base, 80 at the crown (layer 2 ends).
5. Rebar pocket Ø55 in the leaf, bar Ø12 every ≈ 600 along the path, grouted after placing.
6. Starter bars Ø12 from the ring beam into the pockets, lap 650; roof bars stop where the load allows (open).
7. Ring bars Ø12 in the pocket zone every 300 (assumed).
8. Ring beam 400 x 500, cast in place: 6 Ø16, stirrups Ø8 / 150; embedded in the buttress notch.
9. Buttress 400 wide, 700 from the inner face (300 beyond the wall), L-shaped around the ring beam; top flush with the roof base at +2.50.
10. Wall 400 printed, same pattern as the roof; Ø16 bars in the pockets run from the footing into the beam and on into the roof pockets (assumed).

● from your sketches
○ assumed by me, to confirm

Notes

Layer pattern from your drawing: from the inside, 80 + 80 printed layers (the second curves round the bar), then a 160 zig-zag zone, then the 80 outer layer; 400 in total.

Wall and roof are 3D printed in 10 mm layers (horizontal joints drawn every 10). Roof thicknesses are square to the slope; the wall is vertical. Black heavy lines and dots = rebar, white lattice cells = open air, thin horizontal lines = printed layers, single hatch = other concrete, crossed hatch = ring beam cut in the section plane.

0 1 2 3 4 5 dm

Scale 1:10 at A3 · dimensions in mm · draft 1, 2026-10-11
pyramid-roof · hand estimates, not engineered